

3(a)	I take antibiotics any two from: 1 wash hands, qualified ; e.g. thoroughly / regularly / with soap before handling food <i>context is, preparing / eating</i> I cooking after, urination / defaecation A after using the toilet / AW I properly / correctly 2 cook food, qualified ; e.g. thoroughly / completely / to destroy pathogen ; A ref. to uncontaminated food e.g. from commercial (safe) sources bod boiling food 3 drink water that is, safe / bottled / treated / chlorinated / uncontaminated / boiled / filtered to remove pathogens / UV-treated ; bod sterilised / piped A avoid bottle feeding if water source unknown I clean / sanitised / pure R disinfected R drink boiling water 4 do not, buy / use, ice from untrusted sources ; AW 5 do not eat (raw), fruit / vegetables, washed in water from unknown sources / prepared by other people / <i>ref. to</i> plants irrigated with contaminated water ; 6 avoid eating, (potentially contaminated) raw fish / shellfish ; 7 do not, swim in (potentially) / wash in, (potentially) contaminated water ; 8 AVP ; e.g. do not share towels cover food from flies	2
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Question	Answer	Marks															
3(b)(i)	<i>candidates should use evidence that does not agree with statement 'as the number of cases of cholera decreases, the CFR also decreases'</i> correct calculation of missing CFR ; <i>allow written into Table 3.1</i> Democratic Republic of the Congo (DRC) 0.9 (%) R 0.898 (%) or Sudan 2.8 (%) bod 2.77 (%) correct evidence referencing DRC or Sudan ; e.g. DRC highest number of cases but lower CFR than, Malawi / Ethiopia / Sudan R Somalia Sudan has lowest cases but highest CFR or higher than, DRC / Malawi / Ethiopia / Somalia A use of relevant data (from calculation and Table 3.1) <table border="1"> <tr> <td><i>DRC</i></td><td><i>52 654</i></td><td><i>0.9</i></td></tr> <tr> <td><i>Malawi</i></td><td><i>32 530</i></td><td><i>2.5</i></td></tr> <tr> <td><i>Ethiopia</i></td><td><i>30 389</i></td><td><i>1.5</i></td></tr> <tr> <td><i>Somalia</i></td><td><i>18 304</i></td><td><i>0.3</i></td></tr> <tr> <td><i>Sudan</i></td><td><i>8 895</i></td><td><i>2.8</i></td></tr> </table> <i>'agree', or incorrect CFRs calculated: allow one mark ecf for quoting relevant comparative data from Table 1.1 and (if stated) the incorrect values must include, country / number of cases, <u>and</u> CFR value</i>	<i>DRC</i>	<i>52 654</i>	<i>0.9</i>	<i>Malawi</i>	<i>32 530</i>	<i>2.5</i>	<i>Ethiopia</i>	<i>30 389</i>	<i>1.5</i>	<i>Somalia</i>	<i>18 304</i>	<i>0.3</i>	<i>Sudan</i>	<i>8 895</i>	<i>2.8</i>	2
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3(b)(ii)	any three from: <i>assume points about Malawi if not stated, ora for Ethiopia</i> 1 ref. to medical treatment, qualified ; e.g. fewer qualified medical personnel fewer, health centres / hospitals less life-saving equipment available 2 antibiotics given were, not effective / used against drug-resistant strains ; AW 3 antibiotics (for severe cases), qualified ; e.g. did not receive / example – could not afford / lack of supply from authorities received / treated / diagnosed, too late 4 oral rehydration, salts / therapy, qualified ; A ORT / ORS e.g. insufficient / no access to, oral rehydration points / ORPs <i>allow examples given in mp3</i> 5 late diagnosis / long time before treatment began ; <i>allow if not stated for mp3 or 4</i> 6 people with cholera have no access to, safe / uncontaminated, drinking water or ref. to contaminated water used to make up ORS ; 7 (initial) misdiagnosis / false-negative test result (so, no / incorrect, treatment) ; 8 <i>idea of less (targeted) education about treatment (of cholera) / did not seek treatment ; I ref. to prevention</i> <i>'more people were educated to get oral rehydration salts for treatment' is mp4 and mp8</i> 9,10 AVP ;; e.g. country / areas affected, proportionately higher people at risk e.g. elderly / infants / with other conditions more accurate reporting of numbers of death ref. to inaccessibility owing to, conflict / natural disasters <i>idea that greater proportion of people with cholera also had HIV/AIDs</i>	3

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3(c)(i)	<i>ref. to part of DNA not needed as this is in the question</i> sequence of, bases / nucleotides ; coding for a polypeptide ; A protein <i>for polypeptide</i> bod expressing <i>for coding</i> A <i>idea of</i> genetic information for the synthesis of a (specific) protein	2
3(c)(ii)	any two from: I plasmids 70S ribosomes ; cell wall made of, peptidoglycan / murein ; AVP ;; <i>allow other non-syllabus bacterial structures found in V. cholerae</i> e.g. capsule flagellum made of flagellin pilus / pili	2